

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – 3.2 LINEAR MOTION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the Driving Question: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes? Work on your own. Write your answers in the space under each prompt. Use what you observed, measured and calculated in Lessons 1 to 7. Show every calculation: write the known values with units, name the equation you choose, and give your final answer with units. Use $g = 10 \text{ m/s}^2$ in all calculations. Ignore friction and air resistance unless the question says otherwise. Remember Mzee Kamau's mkokoteni (300 kg when loaded) is released from rest on a straight, steady slope on the Limuru-Nairobi escarpment road where $\sin(\text{angle}) = 0.2$. It crashes into a parked lorry after 100 m along the road.

Part 1: Explaining what Achieng saw

Achieng, a bystander, said: 'It started slowly then flew.' Mzee Kamau's mkokoteni has no engine and nobody was pushing it. Write a Claim, Evidence, Reasoning explanation of why the cart keeps getting faster. In your answer:

- (a) State your claim.**
- (b) Use at least three values from the cart's motion (for example, its speed at different times) as evidence.**
- (c) Explain why the cart started slowly, and why the crash felt so sudden, using the idea of how distance changes with time.**
- (d) Say whether the cart's motion is free fall, and explain your answer.**

Claim: The mkokoteni keeps getting faster because gravity pulls it down the slope all the time. This gives it a constant acceleration of 2.0 m/s^2 ($a = g \times 0.2 = 10 \times 0.2$), so its velocity increases steadily even though nobody pushes it and it has no engine.

Evidence: Released from rest, the cart's velocity rises by 2 m/s every second. It is 4 m/s after 2 s , 10 m/s after 5 s and 20 m/s (72 km/h) after 10 s , when it has travelled 100 m and hits the lorry. The distance is 25 m by 5 s but 100 m by 10 s .

Reasoning: Acceleration is the change in velocity per unit time. Because the force of gravity down the slope acts all the time, the acceleration stays constant and the velocity keeps growing by 2 m/s each second. The cart started slowly because its velocity began at 0 m/s , so it covered only 1 m in the first second. The crash felt sudden because with $u = 0$ the distance grows as $s = t^2$. The cart covers 25 m in the first 5 s but 75 m more in the next 5 s , so the last seconds were very fast. The cart is not in free fall. Free fall has $a = g = 10 \text{ m/s}^2$ on a vertical path. The slope gives the cart only 0.2 of g , so its acceleration is 2 m/s^2 .

Note: Achieng's words 'started slowly then flew' match this. The speed starts at zero and grows every second, and the distance covered in each second grows too.

Part 2: Predicting the crash using the equations of linear motion

The cart is released from rest ($u = 0$) with $a = 2.0 \text{ m/s}^2$ and crashes after travelling 100 m along the slope.

(a) List the five variables used in the equations of linear motion and the knowns and the unknown for finding the speed at the crash.

(b) Choose one of the four equations of linear motion, explain in words why it is the right one, and calculate the speed at the crash in m/s and in km/h.

(c) Use a different equation to find the time taken to reach the lorry, and show that it agrees with the data from the lessons.

(d) Find the speed of the cart after it has travelled 25 m. Then compare 25 m with 100 m: by what factor has the distance changed, by what factor has the speed changed, and what does this tell you about how speed depends on distance?

(a) The five variables are u (initial velocity, m/s), v (final velocity, m/s), a (acceleration, m/s^2), s (displacement, m) and t (time, s). Knowns: $u = 0 \text{ m/s}$, $a = 2.0 \text{ m/s}^2$, $s = 100 \text{ m}$. Unknown: v . (t is not given and is not needed.)

(b) The equation that contains u , a , s and v , but not t , is $v^2 = u^2 + 2as$. $v^2 = 0 + 2 \times 2.0 \times 100 = 400 \text{ m}^2/\text{s}^2$, so $v = 20 \text{ m/s}$. In km/h: $20 \times 3.6 = 72 \text{ km/h}$.

(c) Use $v = u + at$: $20 = 0 + 2.0 \times t$, so $t = 20 \div 2.0 = 10 \text{ s}$. Check with $s = ut + \frac{1}{2}at^2$: $s = 0 + \frac{1}{2} \times 2.0 \times 10^2 = 100 \text{ m}$. This agrees with the crash at 10 s after 100 m.

(d) With $s = 25 \text{ m}$: $v^2 = 0 + 2 \times 2.0 \times 25 = 100$, so $v = 10 \text{ m/s}$. Going from 25 m to 100 m, the distance becomes four times as large ($100 \div 25 = 4$), while the speed goes from 10 m/s to 20 m/s, which is two times as large. So the speed is not proportional to the distance: the distance must become four times as large for the speed to double (because $v^2 = 2as$, v depends on the square root of s).

Part 3: Free fall compared with the cart

Class drop data: a rubber bung, a stone and a crumpled paper ball were each dropped from $h = 2.0 \text{ m}$. The mean times of 5 trials were 0.63 s, 0.64 s and 0.65 s respectively.

(a) What do these results show about the effect of mass on how objects fall when air resistance is negligible?

(b) Use the class mean time of 0.64 s to estimate g , and explain why we use $g = 10 \text{ m/s}^2$ in

(a) The three objects have very different masses but fell in almost the same time (0.63 s, 0.64 s, 0.65 s), and the small differences are within reaction-time error. So when air resistance is negligible, all objects fall with the same acceleration due to gravity, regardless of mass.

(b) $g = 2h/t^2$ (from $s = \frac{1}{2}at^2$ with $u = 0$) = $2 \times 2.0 \div 0.64^2 = 4 \div 0.4096 = 9.8 \text{ m/s}^2$. We use $g = 10 \text{ m/s}^2$ because it is a convenient value very close to the measured one, and the difference from our result is small enough to be explained by reaction-time error.

(c) Take down as positive, $u = 0$, $a = 10 \text{ m/s}^2$, $s = 20 \text{ m}$. $s = \frac{1}{2}at^2$, so $20 = \frac{1}{2} \times 10 \times t^2$, $t^2 = 4$, $t = 2 \text{ s}$. $v = u + at = 0 + 10 \times 2 = 20 \text{ m/s}$.

(d) The stone and the cart both reach 20 m/s after a 20 m vertical drop, but the stone takes 2 s while the cart takes 10 s. The slope only provides 0.2 of g , so the cart's acceleration is 2 m/s^2 instead of 10 m/s^2 , and it takes five times longer to reach the same

calculations. (c) A stone is dropped from rest through a vertical drop of 20 m, the same vertical drop as the cart's. Calculate the time taken and the speed on reaching the ground. (d) Compare your answers in (c) with the cart's time and speed. Explain the difference in time using the idea of the slope, and say what type of slope free fall represents.	speed. Free fall is the extreme case of a vertical slope ($\sin = 1$), where $a = g$.
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Part 4: Slowing down and the sign convention

A boda-boda rider, Otieno, is travelling at 15 m/s and brakes before a police check with a constant acceleration of -3 m/s^2. (a) Explain what the negative sign tells you, and compare it with the sign of the cart's acceleration down the slope. (b) Calculate how long Otieno takes to stop and how far he travels while stopping. (c) A ball is thrown upward at 15 m/s from a 20 m high Mombasa rooftop. Taking up as positive, state the acceleration of the ball at the top of its path and explain your answer. Then calculate its velocity when it hits the ground.	(a) The negative sign means the acceleration is in the opposite direction to the motion, so the velocity decreases (deceleration). Otieno takes the direction of motion as positive and his acceleration points backwards. The cart's acceleration of $+2.0 \text{ m/s}^2$ is in the same direction as its motion down the slope, so its velocity increases. (b) $u = 15 \text{ m/s}$, $v = 0$, $a = -3 \text{ m/s}^2$. Time: $v = u + at$ gives $0 = 15 - 3t$, so $t = 5 \text{ s}$. Distance: $v^2 = u^2 + 2as$ gives $0 = 225 + 2(-3)s$, so $s = 225 \div 6 = 37.5 \text{ m}$. Check: $s = \frac{1}{2}(u+v)t = \frac{1}{2} \times 15 \times 5 = 37.5 \text{ m}$. (c) The acceleration of the ball is $a = -10 \text{ m/s}^2$ all the time, including at the top, where $v = 0$ for an instant. Gravity keeps acting, so the velocity keeps changing by 10 m/s each second downward. The ball ends at the ground, which is 20 m below the start, so $s = -20 \text{ m}$. Using $v^2 = u^2 + 2as$: $v^2 = 225 + 2(-10)(-20) = 225 + 400 = 625$, so $v = -25 \text{ m/s}$ (the negative root because it moves downward). The ball hits the ground at 25 m/s downward. Check: it takes 4 s, since $v = 15 - 10 \times 4 = -25 \text{ m/s}$.
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Part 5: Road safety and the final explanation

A matatu on Mombasa Road has a driver reaction time of 1.0 s. On a dry road, braking gives a deceleration of 5 m/s^2. On a wet road, the deceleration is 2.5	(a) $72 \text{ km/h} \div 3.6 = 20 \text{ m/s}$. Reaction distance = speed $\times$ reaction time = $20 \times 1.0 = 20 \text{ m}$. Braking distance: $v^2 = u^2 + 2as$ with $v = 0$, so $0 = 400 + 2(-5)s$ and $s = 400 \div 10 = 40 \text{ m}$. Total stopping distance = $20 + 40 = 60 \text{ m}$. (b) Wet road: $s = u^2/2 a = 400 \div (2 \times 2.5) = 400 \div 5 = 80 \text{ m}$. The wet-road braking distance is 80 m, which is double the 40 m on a dry road, because the deceleration is half as large.
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m/s². (a) The matatu travels at 72 km/h. Convert this to m/s, then calculate the reaction distance, the braking distance on a dry road and the total stopping distance. (b) Find the braking distance of the same matatu on a wet road, and compare it with the dry road. (c) The matatu's speed is increased to 108 km/h. Find the new braking distance on a dry road, and say by what factor it has changed compared with part (a) and why. (d) Explain how your answers help us understand why the Kinangop descent has warning signs and why a runaway mkokoteni with no brakes is so dangerous. Suggest TWO ways to reduce the dangers of speed on steep roads.	(c) $108 \text{ km/h} \div 3.6 = 30 \text{ m/s}$. Braking distance = $900 \div 10 = 90 \text{ m}$. This is $90 \div 40 = 2.25$ times the 40 m at 20 m/s. Braking distance depends on the square of speed ($u^2/2 a$), so increasing the speed by a factor of 1.5 multiplies the braking distance by $1.5^2 = 2.25$. (Doubling the speed multiplies the braking distance by 4, and tripling it multiplies it by 9.) (d) The mkokoteni reaches 20 m/s (72 km/h) with no brakes, and a matatu at the same speed needs 60 m to stop on a dry road, so a cart that cannot brake cannot avoid the lorry. Because braking distance grows with the square of the speed, steep roads like the Kinangop descent have warning signs so drivers slow down early. Ways to reduce the danger: (1) drive at a lower speed and brake early on slopes, following NTSA speed and safe-distance rules; (2) keep a larger gap from the vehicle ahead and reduce speed on wet roads. (Other valid answers: apply a handbrake or put a block under a cart's wheels when parked on a slope.)
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the phenomenon (claim, evidence, reasoning)	Gives a clear claim that gravity acts down the slope all the time giving a constant acceleration of 2.0 m/s^2 . Uses at least three correct cart values (such as 4 m/s at 2 s, 10 m/s at 5 s, 20 m/s at 10 s and 100 m). Explains why it starts slowly and why the crash feels sudden ($s = t^2$), and correctly says the cart is not in free fall.	Gives a correct claim with some correct data, and explains acceleration, but explains only one of 'starts slowly' or 'feels sudden', or does not address free fall.	Claim is unclear or only says the cart is 'pulled' or 'pushed' without linking to constant acceleration, with little or no data used.
Using the equations of linear motion	Lists the five variables with units, writes the knowns and unknown, chooses a suitable equation and justifies the choice, calculates correctly with units (20 m/s, 72 km/h, 10 s, 10 m/s at 25 m), verifies with a second equation, and correctly	Chooses suitable equations and gets most answers correct with units, but gives little justification for the choice or does not check with a second equation.	Equations are chosen at random or substituted incorrectly, or answers have missing or wrong units.

	compares how distance and speed change between 25 m and 100 m.		
Free fall and the comparison with the slope	Explains that mass does not change the fall time, estimates g correctly (9.8 m/s^2) and explains the use of 10 m/s^2 . Calculates 2 s and 20 m/s for the 20 m drop and explains that the slope gives only 0.2 of g , with free fall as the extreme case of a vertical slope.	Calculates the free-fall values correctly and says the cart is slower, but the explanation of why (the slope fraction of g) is incomplete.	Does not link results to the effect of mass or to the slope, or uses $a = 10 \text{ m/s}^2$ for the cart.
Sign convention and use of scalar/vector terms	Chooses a positive direction explicitly, interprets negative acceleration as opposite to the motion, correctly uses $a = -10 \text{ m/s}^2$ for the upward-thrown ball including at the top, and gets the 37.5 m, 5 s and -25 m/s answers with correct signs.	Uses signs mostly correctly and gets most answers right, but does not explain the signs or makes one sign error.	Does not use signs consistently, or says the acceleration is zero at the top of the ball's path.
Applying the ideas to road safety	Converts units correctly, calculates reaction, braking and total stopping distances (20 m, 40 m, 60 m; 80 m wet; 90 m at 30 m/s), explains the square relationship, and gives two sensible ways to reduce danger linked to Kinangop signs and NTSA rules.	Calculates most distances correctly and gives sensible safety measures, but explanation of the square relationship is weak.	Makes errors in unit conversion or stopping distance, or gives safety ideas that are not linked to the calculations.

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